MAC (Medium Access Control)sublayer

Network links can be divided into two categories: those using point-to-pointconnections and those using broadcast channels. In any broadcast network, the key issue is how to determine who gets to usethe channel when there is competition for it.
The protocols used to determine who goes next on a multiaccess channel belongto a sublayer of the data link layer called the **MAC** (**Medium Access Control**)sublayer. The MAC sublayer is especially important in LANs, particularlywireless ones because wireless is naturally a broadcast channel.

The Channel Allocation Problem : The channel might be a portion of the wireless spectrumin a geographic region, or a single wire or optical fiber to which multiple nodesare connected. It does not matter. In both cases, the channel connects each user toall other users and any user who makes full use of the channel interferes withother users who also wish to use the channel.

MULTIPLE ACCESS PROTOCOLS

Many algorithms for allocating a multiple access channel are known, some of them are

1. Random Access Protocol
    1. ALOHA
        - Pure ALOHA
        - Slotted ALOHA
    2. Carrier Sense Multiple Access Protocols
    3CSMA with Collision Detection
2. Collision-Free Protocols
    1. A Bit-Map Protocol
    2. Token Passing
    3. Binary Countdown
    4. The Adaptive Tree Walk Protocol
5. Wireless LAN Protocols

ALOHA:

Two versions of ALOHA here: pure and slotted. They differwith respect to whether time is continuous, as in the pure version, or divided intodiscrete slots into which all frames must fit.

Pure ALOHA:

The basic idea of an ALOHA system is simple: let users transmit wheneverthey have data to be sent. There will be collisions, of course, and the collidingframes will be damaged.

If the frame was destroyed, the sender just waits a random amount of time andsends it again. The waiting time must be random or the same frames will collideover and over, in lockstep. Systems in which multiple users share a commonchannel in a way that can lead to conflicts are known as **contention** systems.

A sketch of frame generation in an ALOHA system is given in Fig. 4-1.

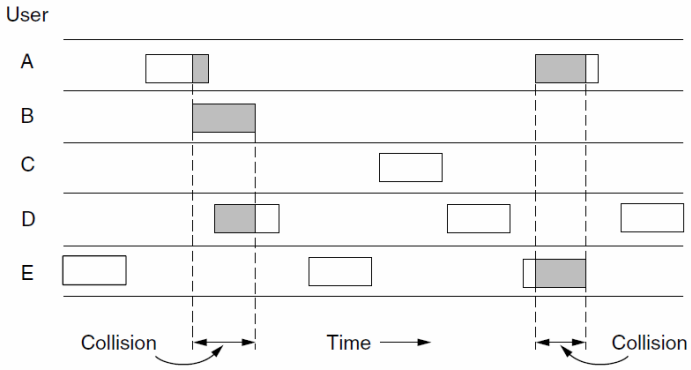
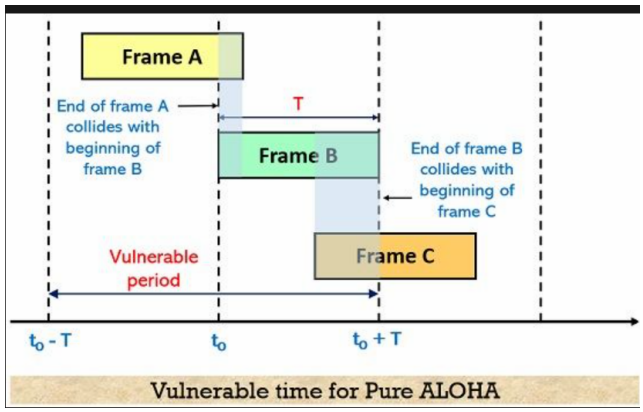


Figure 4-1. In pure ALOHA, frames are transmitted at completely arbitrary times.

Whenever two frames try to occupy the channel at the same time, there willbe a collision (as seen in Fig. 4-1) and both will be garbled. If the first bit of anew frame overlaps with just the last bit of a frame that has almost finished, bothframes will be totally destroyed (i.e., have incorrect checksums) and both willhave to be retransmitted later.

Efficiency of an ALOHA channel (what fraction of all transmitted frames escape collisions under thesechaotic circumstances) :

Let the "frame time" denote the amount of time needed to transmit the standard,fixed-length frame. Station A sends a frame at time *t*. Now imagine station B has already sent a framebetween *t0*- *T*and *t0*. This leads to a collision between the frames from station A andstation B. The end of B's frame collides with the beginning of A's frame. On the otherhand, suppose that station C sends a frame between *t0*and *t0*+ *T* .Here, there is a collisionbetween frames from station A and station C. The beginning of C's frame collideswith the end of A's frame.



The vulnerable time, during which a collisionmay occur in pure ALOHA, is 2 times the frame transmission time.i,e

Pure ALOHA vulnerable time = 2 x T_{fr}

Where T_{fr} is frame tramission time

Throughput of Pure ALOHA:

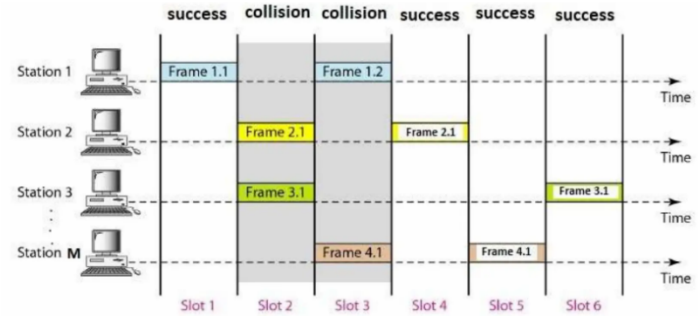
the average number ofsuccessful transmissions for pure ALOHA is

$$S = G \times e^{-2G}$$

The maximum throughputS_{max} is 0.184, for G = 1. In other words, if one-half a frame is generated during one$_2$frame transmission time (in other words, one frame during two frame transmissiontimes), then 18.4 percent of these frames reach their destination successfully. This is anexpected result because the vulnerable time is 2 times the frame transmission time

Slotted ALOHA

In slotted ALOHA we divide the time into slots of T_{fr}s and force the station tosend only at the beginning of the time slot. Figure 12.6 shows an example of framecollisions in slotted ALOHA.



Because a station is allowed to send only at the beginning of the synchronized timeslot, if a station misses this moment, it must wait until the beginning of the next timeslot. This means that the station which started at the beginning of this slot has alreadyfinished sending its frame.

vulnerable time for slotted ALOHA: is one-half that ofpure ALOHA

$$\text{Slotted ALOHA vulnerable time} = T_{fr}$$

Throughput :It can be proved that the average number of successful transmissions forslotted ALOHA is

$$S = G \times e^{-G}.$$

The maximum throughput Smax is 0.368, when G = 1.In other words, if a frame is generated during one frame transmission time, then 36.8percent of these frames reach their destination successfully.

Carrier Sense Multiple Access Protocols:

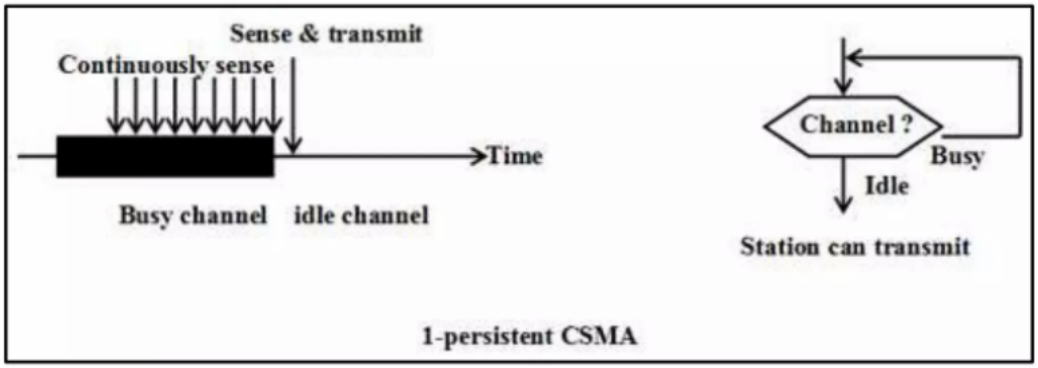
In ALOHA with stations transmitting at will, withoutknowing what the other stations are doing there are bound to be many collisions.In LANs, however, it is often possible for stations to detect what other stationsare doing, and thus adapt their behavior accordingly.

Protocols in which stations listen for a carrier (i.e., a transmission) and act accordingly are called **carrier sense protocols**.A number of them have been proposed , some of them are:

1. **1-persistentCSMA** (**Carrier Sense Multiple Access**).
2. **nonpersistent CSMA**
3. **p-persistent CSMA**

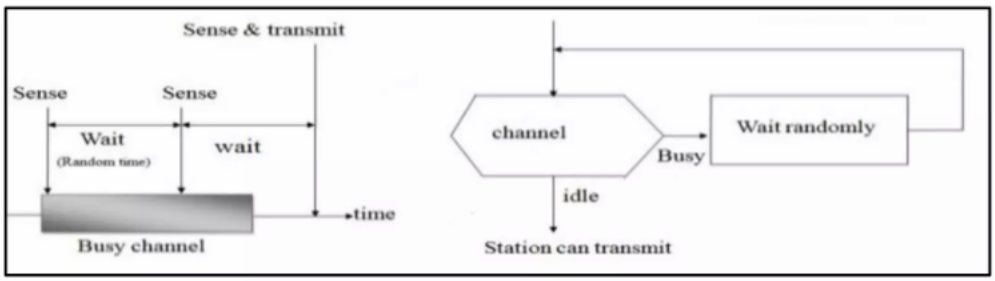
1-persistentCSMA (**Carrier Sense Multiple Access**).

It is the simplestCSMA scheme. When a station has data to send, it first listens to the channelto see if anyone else is transmitting at that moment. If the channel is idle, thestations sends its data. Otherwise, if the channel is busy, the station just waitsuntil it becomes idle. Then the station transmits a frame. If a collision occurs, thestation waits a random amount of time and starts all over again. The protocol iscalled 1-persistent because the station transmits with a probability of 1 when itfinds the channel idle.



nonpersistent CSMA.

In this protocol, aconscious attempt is made to be less greedy than in the previous one. As before, a
station senses the channel when it wants to send a frame, and if no one else issending, the station begins doing so itself. However, if the channel is already inuse, the station does not continually sense it for the purpose of seizing it immediatelyupon detecting the end of the previous transmission. Instead, it waits arandom period of time and then repeats the algorithm.



p-persistent CSMA.

It applies to slotted channels andworks as follows. When a station becomes ready to send, it senses the channel. If it is idle, it transmits with a probability p. With a probability $q = 1 - p$, it defersuntil the next slot. If that slot is also idle, it either transmits or defers again, withprobabilities p and q. This process is repeated until either the frame has beentransmitted or another station has begun transmitting. In the latter case, theunlucky station acts as if there had been a collision (i.e., it waits a random timeand starts again). If the station initially senses that the channel is busy, it waitsuntil the next slot and applies the above algorithm. IEEE 802.11 uses a refinementof p-persistent CSMA.

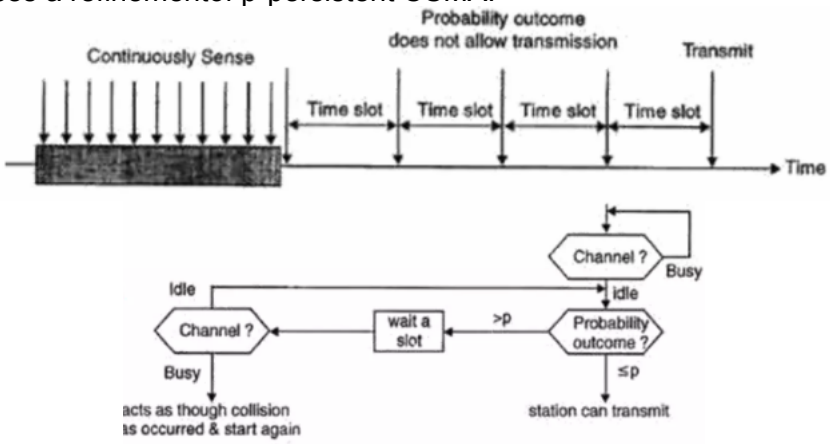


Figure 4-4 shows the computed throughput versus offered traffic for all threeprotocols, as well as for pure and slotted ALOHA.

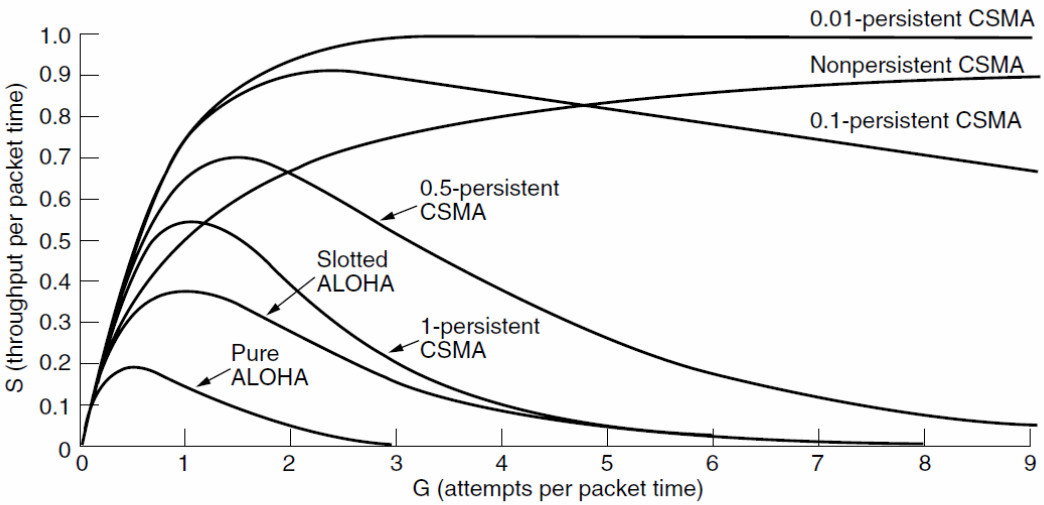


Figure 4-4. Comparison of the channel utilization versus load for various random access protocols.

CSMA with Collision Detection

If two stations sense the channel to be idle and begin transmitting simultaneously, their signals will still collide. Another improvement is for the stations to quickly detect the collision and abruptly stop transmitting,(rather than finishing them) .This protocol, known as **CSMA/CD** (**CSMA with Collision Detection**), is the basis of the classic Ethernet LAN.

If two or more stations decide to transmit simultaneously, there will be a collision. If a station detects a collision, it aborts its transmission, waits a random period of time, and then tries again . Therefore, our model for CSMA/CD will consist of alternating contention and transmission periods, with idle periods occurring when all stations are
Quiet.

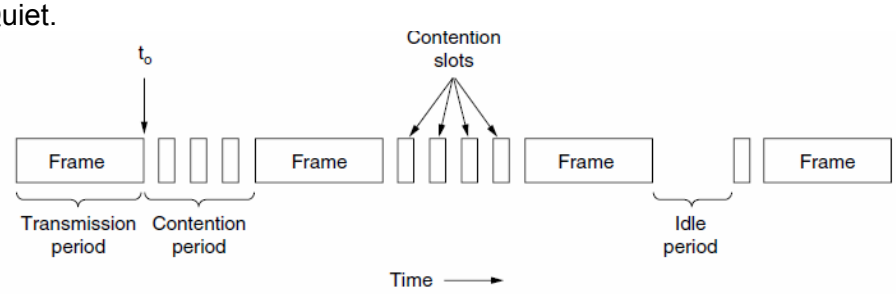


Figure 4-5. CSMA/CD can be in contention, transmission, or idle state.

The minimum time to detect the collision is just the time it takes the signal to propagate from one station to the other

Collision-Free Protocols

In this section, we will examine some protocols that resolve the contention for the channel without any collisions at all, not even during the contention period. Most of these protocols are not currently used in major systems.

1. **A Bit-Map Protocol**

   Bit-Map Protocol is a **contention-based MAC (Medium Access Control) protocol** used to control access to a shared communication channel when multiple stations want to transmit.It solves the problem of **collisions** by giving every station a fixed position in a **bit map**.

   Suppose there are **N stations** in the network. The protocol operates in two phases:

   1. **Contention phase :** There are **N bits**, one for each station.Each station has a fixed bit position. If a station wants to transmit, it sets its corresponding bit to **1**.Otherwise, it leaves the bit as **0**.
   2. **Data transmission phase :** After all stations have indicated whether they want to transmit, the stations transmit **one by one in order**. A station whose bit is 1 gets the opportunity to transmit.

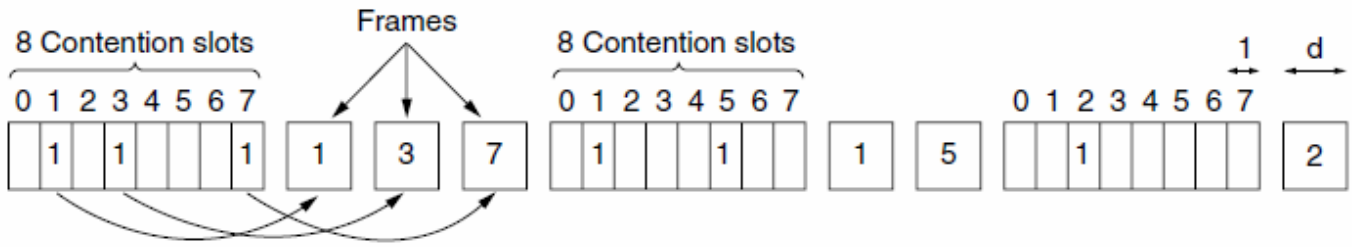


Figure 4-6. The basic bit-map protocol.

2. **Token passing:**

Token Passing is a controlled-access MAC protocol used in computer networks. It allows multiple devices to share a communication medium without collisions. The main idea is that a special small frame called a token circulates among all the devices.

In this methon A token is passed from one station to the next. A station that receives the token checks whether it has data to send.If it has data, it uses the token to transmit. After finishing transmission, it releases/passes the token to the next station. If it has no data, it simply passes the token**.**

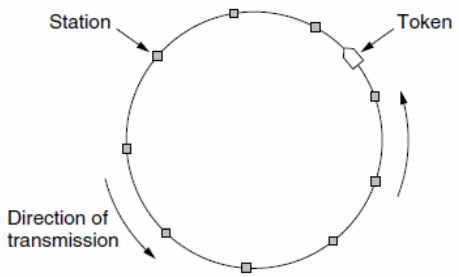
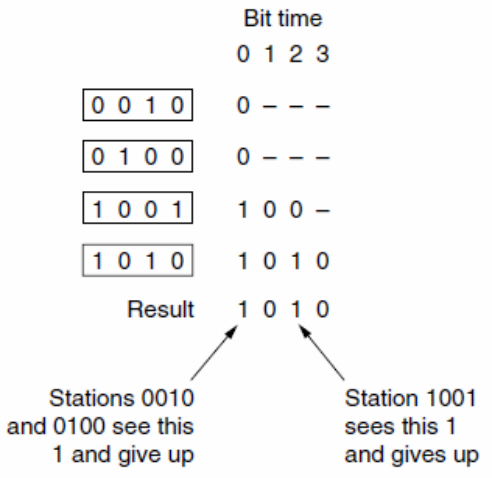


Figure 4-7. Token ring.

3. Binary Countdown Protocol

Binary Countdown is a controlled-access MAC protocol used in computer networks to decide which station gets access to a shared communication channel**.** It uses the binary addresses (IDs) of stations to determine the winner, so there is no collision during data transmission. Each station that wants to transmit participates in a **binary countdown**.

In this method Every station has a unique binary address. Stations that want to transmit place their address on the shared channel, starting from the most significant bit (MSB)**.**If a station sends 0 but detects a 1 on the channel, it knows that another station has a higher-priority address. That station withdraws from the competition. The process continues bit by bit. The station with the highest binary address wins and gets permission to transmit.



Advantages : No collision during data transmission. Efficient when many stations are competing.

Disadvantages : Stations need unique addresses. The station with the highest address gets priority, which can cause unfairness if the same stations repeatedly have data to send. Requires additional control bits before actual data transmission.